ALGEBRAIC SIMPLIFICATION

Factor and reduce, keep the restriction

Simplification makes evaluation easier and reveals structure, but it must respect the domain.

01

QUOTIENT — BEFORE

$$\frac{x^2 - 9}{x + 3}$$

↓ Factor the numerator — $x^2 - 9 = (x - 3)(x + 3)$

02

FACTORED

$$\frac{(x - 3)(x + 3)}{x + 3}$$

↓ Reduce — cancel the common factor $x + 3$, and record $x \neq -3$

03

SIMPLIFIED FORM

AFTER

$$x - 3 \quad (x \neq -3)$$

THE HOLE